

ONLINE RETAIL BUSINESS INTELLIGENCE & ANALYTICS

Project Report

Data Analytics with AI Internship – Aug 2026

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Repository: Siddhib2004/IBM_Siddhi_Online_Retail_Project

1. Executive Summary

This project implements an end-to-end Business Intelligence and Analytics solution for online retail transaction data. The application moves from raw data ingestion and quality checks to descriptive, diagnostic, predictive and prescriptive analytics, and presents the results through an interactive Streamlit dashboard.

The generated outputs cover 13 monthly periods from December 2010 through December 2011. Across those monthly outputs, recorded revenue totals approximately **£10,642,110.80**, with **19,960** monthly order counts. The highest monthly revenue occurs in **Nov-2011** at **£1,503,866.78**.

2. Project Objectives

- Build a reusable pipeline for loading and cleaning Online Retail transaction data.
- Calculate retail KPIs including revenue, orders, customers, products, units and AOV.
- Analyze monthly trends, products and geographic markets.
- Perform diagnostic analysis for cancellations and concentration.
- Forecast the next three months of revenue using Random Forest regression.
- Generate business-oriented recommendations.
- Expose the workflow through Streamlit and CSV outputs.

3. Dataset and Data Source

The project uses the UCI Machine Learning Repository Online Retail dataset. The repository identifies the dataset as UCI Online Retail and includes Online Retail.xlsx. The transaction schema includes invoice number, stock code, description, quantity, invoice date, unit price, customer ID and country.

4. Technology Stack

Technology	Purpose
Python	Core implementation
Pandas	Cleaning and aggregation
NumPy	Numerical calculations
OpenPyXL	Excel input
Plotly	Interactive charts
Streamlit	Dashboard
Scikit-learn	Random Forest forecasting

5. Analytical Workflow

Raw Excel Data → Cleaning → KPI Layer → Descriptive Analysis → Diagnostic Analysis → Predictive Forecast → Prescriptive Recommendations → Dashboard/CSV Outputs

The main implementation is contained in project.py. It loads the workbook, cleans records, computes business metrics, creates analysis tables, trains the forecasting model, generates recommendations, saves outputs and renders the Streamlit interface.

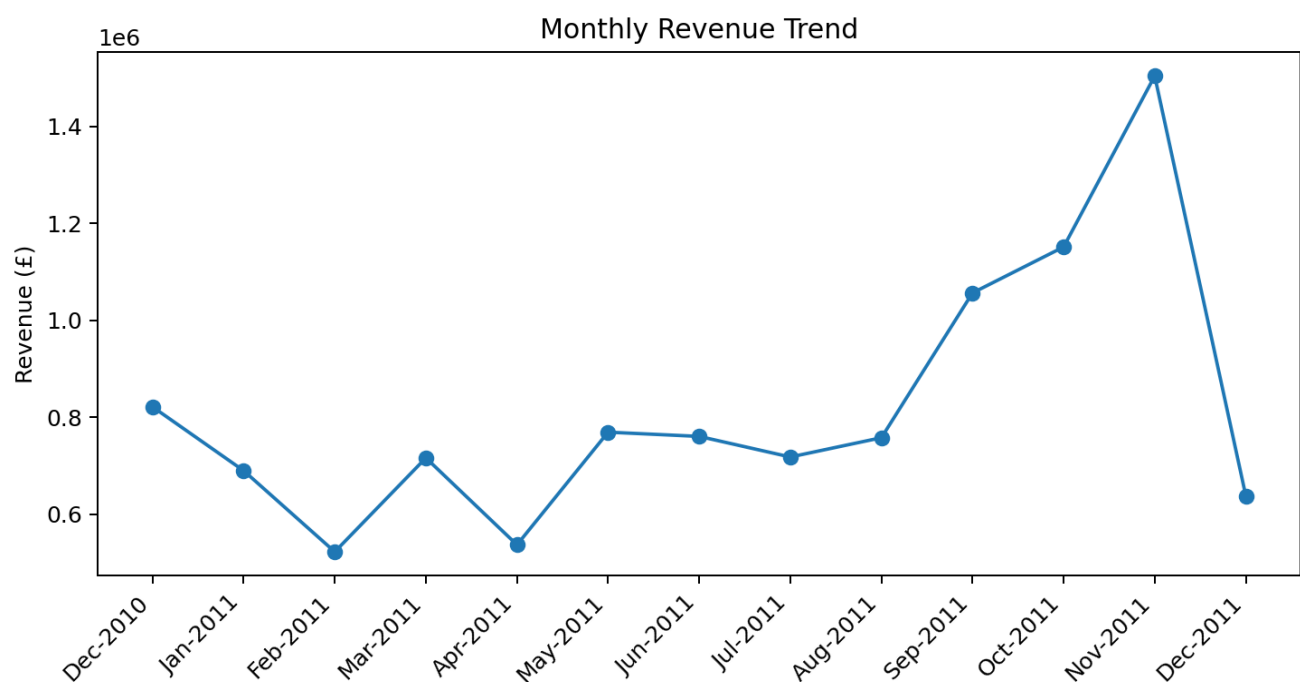
6. Data Cleaning and Feature Engineering

- Remove empty rows and duplicates.

- Convert dates and numeric fields to appropriate types.
- Trim text fields.
- Flag cancellation invoices using invoice numbers beginning with C.
- Exclude invalid records from sales analysis.
- Retain positive, non-cancelled sales for revenue calculations.
- Create Revenue = Quantity × UnitPrice.
- Create Year, Month, YearMonth, Day, DayName and Hour features.

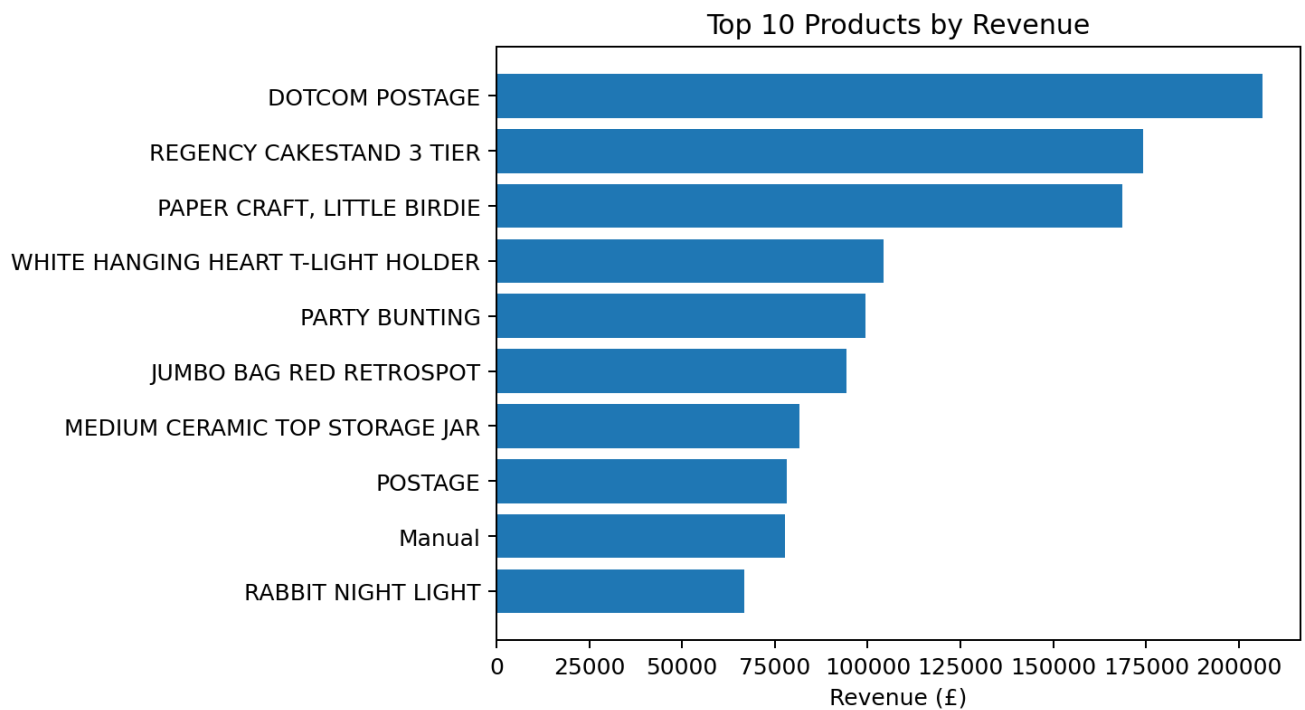
7. KPI and Descriptive Analysis

The repository output indicates total monthly revenue of approximately **£10,642,110.80**. The peak month is **Nov-2011** with **£1,503,866.78**. Revenue in December 2011 changed by **-57.59%** compared with November 2011.

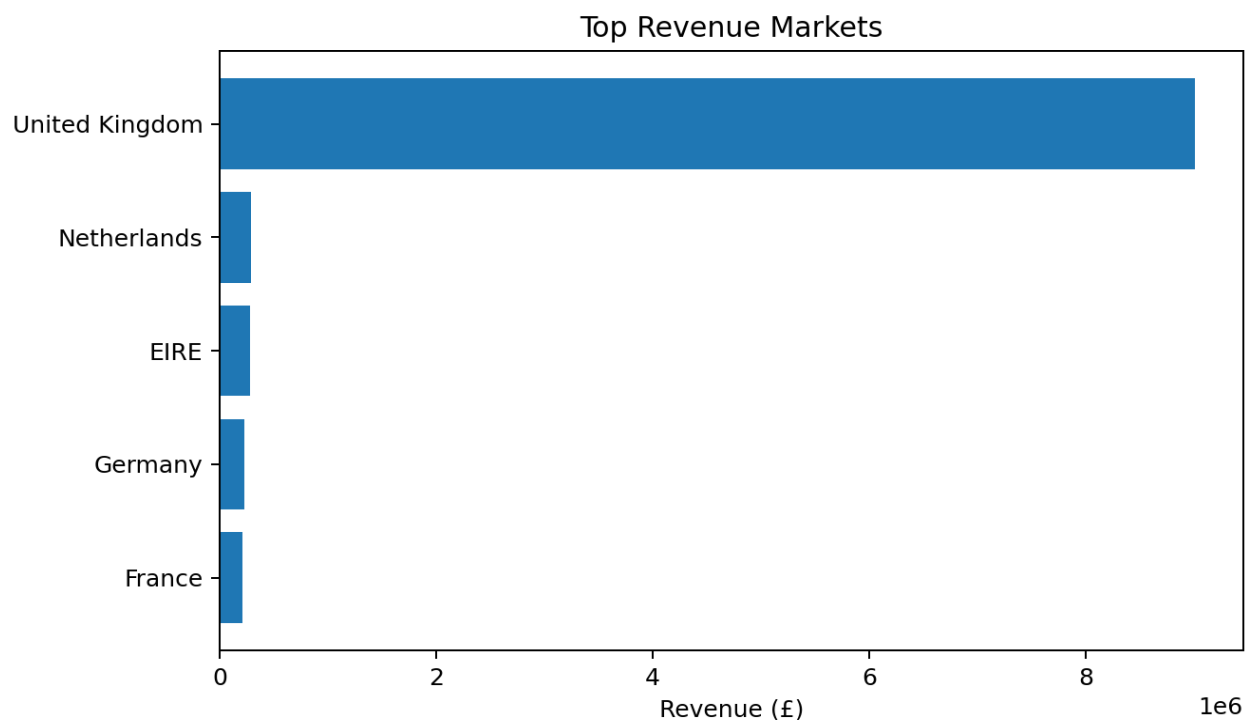


8. Product and Geographic Analysis

The product output ranks products by revenue, units and order count. The leading revenue entries include DOTCOM POSTAGE, REGENCY CAKESTAND 3 TIER, PAPER CRAFT, LITTLE BIRDIE, WHITE HANGING HEART T-LIGHT HOLDER and PARTY BUNTING.



The country output shows the United Kingdom as the largest revenue market at approximately **£9,001,744.09**, followed by the Netherlands, EIRE, Germany and France.



9. Diagnostic Analytics

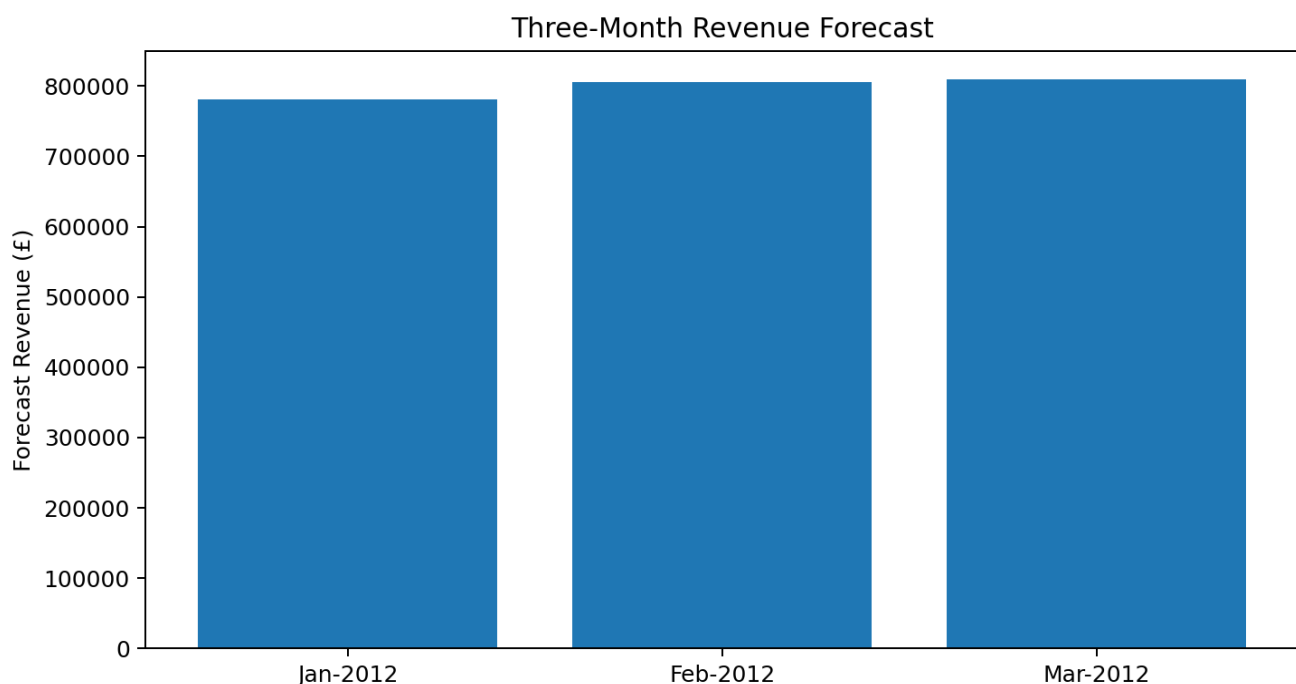
- Cancellation rate analysis for operational risk.
- Product revenue concentration.
- Country revenue concentration.
- Top-10 customer revenue concentration.
- Correlation analysis among quantity, unit price and revenue.

10. Predictive Analytics – Revenue Forecasting

The model uses RandomForestRegressor with year, month, one-, two- and three-month revenue lags and a three-month rolling mean. The implementation uses an 80/20 time-ordered split and evaluates MAE, RMSE and R².

- 300 trees
- random_state = 42
- max_depth = 8
- min_samples_leaf = 2
- forecast horizon = 3 months

The repository forecast output gives approximately **£780,277.00** for January 2012, **£805,949.26** for February 2012 and **£808,924.58** for March 2012.



11. Prescriptive Analytics and Business Insights

- Prioritize inventory and visibility for high-revenue products.
- Review low-revenue products for bundling, pricing and assortment decisions.
- Use geographic revenue patterns to support market-specific retention and localization analysis.
- Monitor cancellations by month and product.
- Use customer concentration analysis for retention and cross-selling.
- Investigate the December 2011 decline before using historical growth rates for planning.
- Treat forecasts as planning signals rather than guarantees.

12. Interactive Dashboard

The Streamlit application provides Executive Dashboard, Data Quality & Cleaning, Descriptive Analysis, Diagnostic Analysis, Predictive Analysis, Prescriptive Analysis, Detailed Tables and Project Methodology pages. It also supports Excel file upload through the sidebar.

13. Output Files

- cleaned_sales_data.csv
- monthly_kpis.csv
- product_analysis.csv
- country_analysis.csv
- revenue_forecast.csv

14. Limitations

- Forecasting uses historical transaction information only.
- Promotions, holidays, inventory constraints, price changes and external economic factors are not explicitly modeled.
- The monthly time series is relatively short for complex forecasting.
- Results depend on the quality and coverage of the supplied transaction dataset.

15. Future Enhancements

- RFM segmentation and cohort retention analysis.
- Basket affinity and product-category analysis.
- Model comparison with additional forecasting methods.
- Forecast confidence intervals and monitoring.
- More interactive filters and drill-downs.
- Cloud deployment with controlled data upload and authentication.

16. Conclusion

The project demonstrates a complete retail analytics workflow combining data engineering, business intelligence, visualization and machine learning. It transforms transaction-level data into KPIs, trend analysis, product and country analysis, diagnostic indicators, a three-month revenue forecast and business actions.

17. References

1. Siddhib2004, IBM_Siddhi_Online_Retail_Project, GitHub.
2. UCI Machine Learning Repository, Online Retail dataset.
3. Repository source files: project.py, requirements.txt and generated outputs.